

Algebraic Geometry – Exercise Sheet 3

Week 3: Schemes, morphisms, gluing and first global properties
Monday and Thursday afternoon sessions; Lectures 11–15

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lectures 1–11

Exercise 1. Global functions as morphisms to the affine line

★ **CORE**

Let X be a scheme.

- (i) Explain why a global section $s \in \Gamma(X, \mathcal{O}_X)$ determines a morphism

$$f_s : X \longrightarrow \mathbb{A}_{\mathbb{Z}}^1.$$

- (ii) Show that

$$\mathrm{Hom}(X, \mathbb{A}_{\mathbb{Z}}^1) \simeq \Gamma(X, \mathcal{O}_X).$$

- (iii) If $X = \mathrm{Spec}(A)$, describe the morphism associated with $a \in A$.

Exercise 2. The reduced point and the double point

★ **CORE**

Inside $\mathbb{A}_{\mathbb{k}}^1 = \mathrm{Spec}(\mathbb{k}[t])$, consider

$$P = \mathrm{Spec}(\mathbb{k}[t]/(t)), \quad D = \mathrm{Spec}(\mathbb{k}[t]/(t^2)).$$

- (i) Show that the two closed immersions have the same image as subsets of $\mathbb{A}_{\mathbb{k}}^1$.
(ii) Show that P and D are not isomorphic as schemes.
(iii) Compute the maps on local rings at their unique points.

Exercise 3. A morphism between affine schemes

★ **CORE**

Let

$$\varphi : \mathbb{k}[u, v] \longrightarrow \mathbb{k}[t], \quad u \longmapsto t, \quad v \longmapsto t^2.$$

- (i) Describe the induced morphism $f : \mathbb{A}_{\mathbb{k}}^1 \rightarrow \mathbb{A}_{\mathbb{k}}^2$.
(ii) Determine $\ker(\varphi)$ and factor f through a closed subscheme of $\mathbb{A}_{\mathbb{k}}^2$.
(iii) Show that the resulting morphism onto that closed subscheme is an isomorphism.

Exercise 4. An arithmetic spectrum

★★ **CHOICE**

Let $A = \mathbb{Z}[s]/(s^2 - 3)$. Describe the inverse image in $\mathrm{Spec}(A)$ of the points (2) , (3) , and (p) for $p \neq 2, 3$, according to the factorisation of $s^2 - 3$ modulo p .

Thursday afternoon — material available: Lectures 1–14

Exercise 5. Dual numbers and tangent vectors

★★ **CORE**

Let $D = \mathrm{Spec}(\mathbb{k}[\varepsilon]/(\varepsilon^2))$ and let $p = (a, b) \in \mathbb{A}_{\mathbb{k}}^2(\mathbb{k})$.

- (i) Show that morphisms $D \rightarrow \mathbb{A}_{\mathbb{k}}^2$ whose closed point maps to p are uniquely of the form

$$x \longmapsto a + u\varepsilon, \quad y \longmapsto b + v\varepsilon.$$

- (ii) Let $C = \mathbb{W}(f) \subset \mathbb{A}_{\mathbb{k}}^2$. Show that such a morphism factors through C precisely when

$$f(a, b) = 0, \quad f_x(a, b)u + f_y(a, b)v = 0.$$

- (iii) Apply this to the parabola $y = x^2$.

Exercise 6. The line with two origins★★ **CORE**

Glue two copies

$$X_1 = \operatorname{Spec}(\mathbb{k}[t_1]), \quad X_2 = \operatorname{Spec}(\mathbb{k}[t_2])$$

along $D(t_1) \simeq D(t_2)$ by $t_1 \mapsto t_2$. Let X be the resulting scheme.

- (i) Describe the underlying topological space.
- (ii) Show that $\Gamma(X, \mathcal{O}_X) \simeq \mathbb{k}[t]$.
- (iii) Show that the canonical map $X \rightarrow \operatorname{Spec} \Gamma(X, \mathcal{O}_X)$ identifies the two origins.
- (iv) Deduce that X is not affine.

Exercise 7. Global functions on the projective line★★ **CHOICE**

Construct $\mathbb{P}_{\mathbb{k}}^1$ by gluing $U_0 = \operatorname{Spec}(\mathbb{k}[t])$ and $U_1 = \operatorname{Spec}(\mathbb{k}[s])$ along $s = t^{-1}$. Prove that

$$\Gamma(\mathbb{P}_{\mathbb{k}}^1, \mathcal{O}_{\mathbb{P}^1}) = \mathbb{k}.$$

Exercise 8. The projective conic from its standard opens★★ **CORE**

Let

$$S = \mathbb{k}[x_0, x_1, x_2]/(x_0x_2 - x_1^2), \quad C = \operatorname{Proj}(S).$$

- (i) Compute $(S_{x_0})_0$ and $(S_{x_2})_0$.
- (ii) Identify $D_+(x_0)$ and $D_+(x_2)$ with affine lines.
- (iii) Describe their overlap and recover the gluing rule for $\mathbb{P}^1$.

Exercise 9. Connectedness and idempotents★★ **CORE**

- (i) Prove that $\operatorname{Spec}(A)$ is disconnected if and only if A has a non-trivial idempotent.
- (ii) Apply this to $A = \mathbb{k}[t]/(t^2 - t)$.
- (iii) Show that $\operatorname{Spec}(\mathbb{k}[x, y]/(xy))$ is connected although reducible.

Exercise 10. Reduction and its universal property★★ **CORE**

Let

$$A = \mathbb{k}[x, y]/(x^2, xy), \quad X = \operatorname{Spec}(A).$$

- (i) Compute the nilradical and X_{red} .
- (ii) Let R be a reduced $\mathbb{k}$ -algebra. Show that every homomorphism $A \rightarrow R$ factors uniquely through A_{red} .
- (iii) Translate this into a statement about morphisms of schemes.

Exercise 11. A weighted projective chart★★★ **OPTIONAL**

Let $R = \mathbb{k}[x, y, z]$ with $\deg x = \deg y = 1$ and $\deg z = 2$, and set $X = \operatorname{Proj}(R)$. Show that $D_+(x)$ and $D_+(y)$ are affine planes and that

$$D_+(z) \simeq \operatorname{Spec}(\mathbb{k}[a, b, c]/(ac - b^2)).$$

Identify the singular point of this chart.

After Friday's lecture or preparation for Week 4**Exercise 12. Components and generic points**★★ **CHOICE**

Let

$$I = (xy, xz) \subset \mathbb{k}[x, y, z], \quad X = \operatorname{Spec}(\mathbb{k}[x, y, z]/I).$$

Show that $I = (x) \cap (y, z)$. Determine the irreducible components, their generic points and dimensions, and decide whether X is connected or integral.

Exercise 13. The function field of the cusp★★ **CHOICE**

Let

$$C = \operatorname{Spec}(\mathbb{k}[x, y]/(y^2 - x^3)).$$

Show that C is integral and that $K(C) \simeq \mathbb{k}(t)$.